

NYCU Introduction to Machine Learning, Homework 2

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Part. 1, Coding (60%):

(25%) Logistic Regression w/ Gradient Descent Method

- (5%) Show the hyperparameters (learning rate and iteration, etc) that you used and the weights and intercept of your model.

```
LR = LogisticRegression(  
    learning_rate=2e-4, # You can modify  
    num_iterations=10000, # You can modify  
)
```

```
| INFO | __main__:main:179 - LR: Weights: [-0.02876863  0.00648972  0.02775907  0.02291548  0.0158755 ], Intercep: -0.09114233910685697
```

- (5%) Show the AUC of the classification results on the testing set.

```
| INFO | __main__:main:180 - LR: Accuracy=0.8095, AUC=0.8455
```

- (15%) Show the accuracy score of your model on the testing set

```
| INFO | __main__:main:180 - LR: Accuracy=0.8095, AUC=0.8455
```

(25%) Fisher Linear Discriminant, FLD

- (5%) Show the mean vectors m_i ($i=0, 1$) of each class, the within-class scatter matrix S_W , and the between-class scatter matrix S_B of the training set.

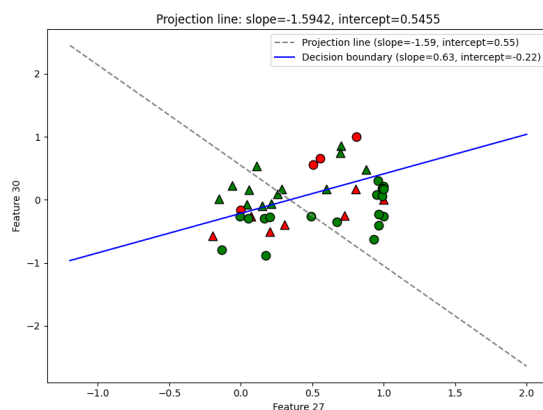
```
2025-10-28 16:24:32.861 | INFO | __main__:main:202 - FLD: m0=[ 0.35994138 -0.04560139], m1=[0.32519126 0.04435118] of cols=['27', '30']  
2025-10-28 16:24:32.861 | INFO | __main__:main:203 - FLD:  
Sw=  
[[41.93041055 15.7202037 ]  
 [15.7202037  37.25186904]]  
2025-10-28 16:24:32.861 | INFO | __main__:main:204 - FLD:  
Sb=  
[[ 0.00120757 -0.00312586]  
 [-0.00312586  0.00809147]]
```

- (5%) Show the Fisher's linear discriminant w of the training set.

```
2025-10-28 16:24:32.861 | INFO | __main__:main:205 - FLD:  
w=  
[-0.00205997  0.00328402]
```

- (15%) Show the accuracy score on the testing set. Also, plot/obtain predictions for the testing set by measuring the distance between the projected value of the testing data and the projected means of the training data for the two classes (Please check the slide for color, shape, and other plotting requirements).

```
2025-10-28 16:24:32.861 | INFO | __main__:main:206 - FLD: Accuracy=0.7381
```



(10%) Code Check and Verification

- (10%) Lint the code and show the PyTest results.

```
(hw2) wuhsinze@HsinZewUWU hw2 % flake8 main.py
(hw2) wuhsinze@HsinZewUWU hw2 %
```

```
(hw2) wuhsinze@HsinZewUWU hw2 % pytest ./test_main.py -s
===== test session starts =====
platform darwin -- Python 3.13.9, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/wuhsinze/Desktop/三上/ML/HW2/hw2
collected 2 items

test_main.py (395, 2) (395,)
2025-10-28 16:37:02.181 | INFO | test_main:test_logistic_regression:35 - accuracy=0.9517
(395, 2) (395,)
2025-10-28 16:37:02.183 | INFO | test_main:test_fld:45 - accuracy=0.8759
+
===== 2 passed in 2.27s =====
```

Part. 2, Questions (40%):

- (15%)

(*) Using (4.57) and (4.58), derive the result (4.65) for the posterior class probability in the two-class generative model with Gaussian densities, and verify the results (4.66) and (4.67) for the parameters $\mathbf{w}$ and w_0 .

$$\begin{aligned} p(C_1|\mathbf{x}) &= \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_1)p(C_1) + p(\mathbf{x}|C_2)p(C_2)} \\ &= \frac{1}{1 + \exp(-a)} = \sigma(a) \end{aligned} \quad (4.57)$$

$$a = \ln \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_2)p(C_2)} \quad (4.58)$$

$$p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + w_0) \quad (4.65)$$

$$\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \quad (4.66)$$

$$w_0 = -\frac{1}{2}\boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{p(C_1)}{p(C_2)}. \quad (4.67)$$

Prove:

$$p(C_1|x) = \frac{1}{1 + \frac{p(x|C_2)p(C_2)}{p(x|C_1)p(C_1)}} = \frac{1}{1 + \exp(\ln(\frac{p(x|C_2)p(C_2)}{p(x|C_1)p(C_1)}))} = \frac{1}{1 + \exp(-\ln(\frac{p(x|C_1)p(C_1)}{p(x|C_2)p(C_2)}))} = \sigma(a)$$

$$p(x | C_k) = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x - \mu_k)^T \Sigma^{-1} (x - \mu_k)\} \rightarrow (4.58)$$

$$a = \ln\left(\frac{p(x|C_1)}{p(x|C_2)}\right) + \ln\left(\frac{p(C_1)}{p(C_2)}\right) = \ln\left(\frac{\frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x - \mu_1)^T \Sigma^{-1} (x - \mu_1)\}}{\frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x - \mu_2)^T \Sigma^{-1} (x - \mu_2)\}}\right) + \ln\left(\frac{p(C_1)}{p(C_2)}\right)$$

$$a = \ln(\exp\{-\frac{1}{2}(x - \mu_1)^T \Sigma^{-1} (x - \mu_1)\}) - \ln(\exp\{-\frac{1}{2}(x - \mu_2)^T \Sigma^{-1} (x - \mu_2)\}) + \ln\left(\frac{p(C_1)}{p(C_2)}\right)$$

$$a = -\frac{1}{2}(x - \mu_1)^T \Sigma^{-1} (x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma^{-1} (x - \mu_2) + \ln\left(\frac{p(C_1)}{p(C_2)}\right)$$

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = x^T \Sigma^{-1} x - 2x^T \Sigma^{-1} \mu + \mu^T \Sigma^{-1} \mu$$

We cancel the same part ($x^T \Sigma^{-1} x$)

$$a = x^T \Sigma^{-1} \mu_1 - \frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 - x^T \Sigma^{-1} \mu_2 + \frac{1}{2} \mu_2^T \Sigma^{-1} \mu_2 + \ln\left(\frac{p(C_1)}{p(C_2)}\right)$$

$$a = x^T (\Sigma^{-1} \mu_1 - \Sigma^{-1} \mu_2) + (-\frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_2^T \Sigma^{-1} \mu_2) + \ln\left(\frac{p(C_1)}{p(C_2)}\right)$$

$$a = (\Sigma^{-1}(\mu_1 - \mu_2))^T x + (-\frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_2^T \Sigma^{-1} \mu_2 + \ln\left(\frac{p(C_1)}{p(C_2)}\right)) = w^T x + w_0$$

$$p(C_1|x) = \sigma(w^T x + w_0)$$

2. (10%)

(a) Give one real-world situation where you would prefer Logistic Regression (LR) over the Perceptron, and explain why.

(b) Is Logistic Regression actually used for regression (predicting a continuous value)? If not, state what task it really solves and why the name includes “regression.”

(a) The bank credit risk assessment

For a regulated industry like finance, where decisions hinge on quantifiable risk metrics and models must be robust to real-world data noise. A bank needs to know a customer's probability of default to calculate the risk-adjusted interest rate and set loan limits, etc. This continuous probability is essential for risk modeling, which LR is a great choice. The Perceptron only gives 1 or 0, couldn't provide necessary degree of certainty. Logistic Regression is the superior choice due to its probabilistic output and guaranteed convergence.

(b) Not really

LR is primarily used for classification tasks, particularly binary ones. More accurately, the model's direct output is a probability representing the likelihood of the input belonging to a specific class. It is called "regression" because it regresses for the log-odds (a continuous value) which are then transformed to estimate the continuous probability of a categorical outcome.

3. (15%)

(a) Why is feature scaling (e.g., standardization or normalization) important in Logistic Regression? Explain two reasons.

(b) If feature scaling is not applied in Logistic Regression, list three problems that may occur. Briefly explain.

(a)

(1) For faster convergence

Typically we would use Gradient Descent for the process, if features have vastly different scales, the cost function forms an elliptical contour, which makes the GD have to take small, zig-zag steps along the valleys, end up very slow. With the scaling, the contours come in more circular shape, can have a more direct path to the minimum, reducing the time.

(2) Ensuring Fair Regularization

In mostly logistic regression, we would apply regularization to prevent overfitting. But we would not want just applying on those with large range, the model might assign a large coefficient to those with small range, then unfairly shrink this towards zero, effectively reducing the influence of it even it is highly predictive. By scaling, we can ensure the coefficients for all features are roughly in same range, making the regularization penalty work equally across all features.

(b)

As partially mentioned in (a), the slow convergence rate and the unfair regularization. Besides, there is another one – difficulty in weight interpretation. A size of a feature's Coefficient is determined by its own scale and unit. A large might simply mean assigned to a small feature, vice versa. This dependency on units makes it hard to compare the coefficients directly to determine the true relative importance of one predictor versus another. Scaling standardizes the basis for comparison, making coefficient interpretation meaningful.